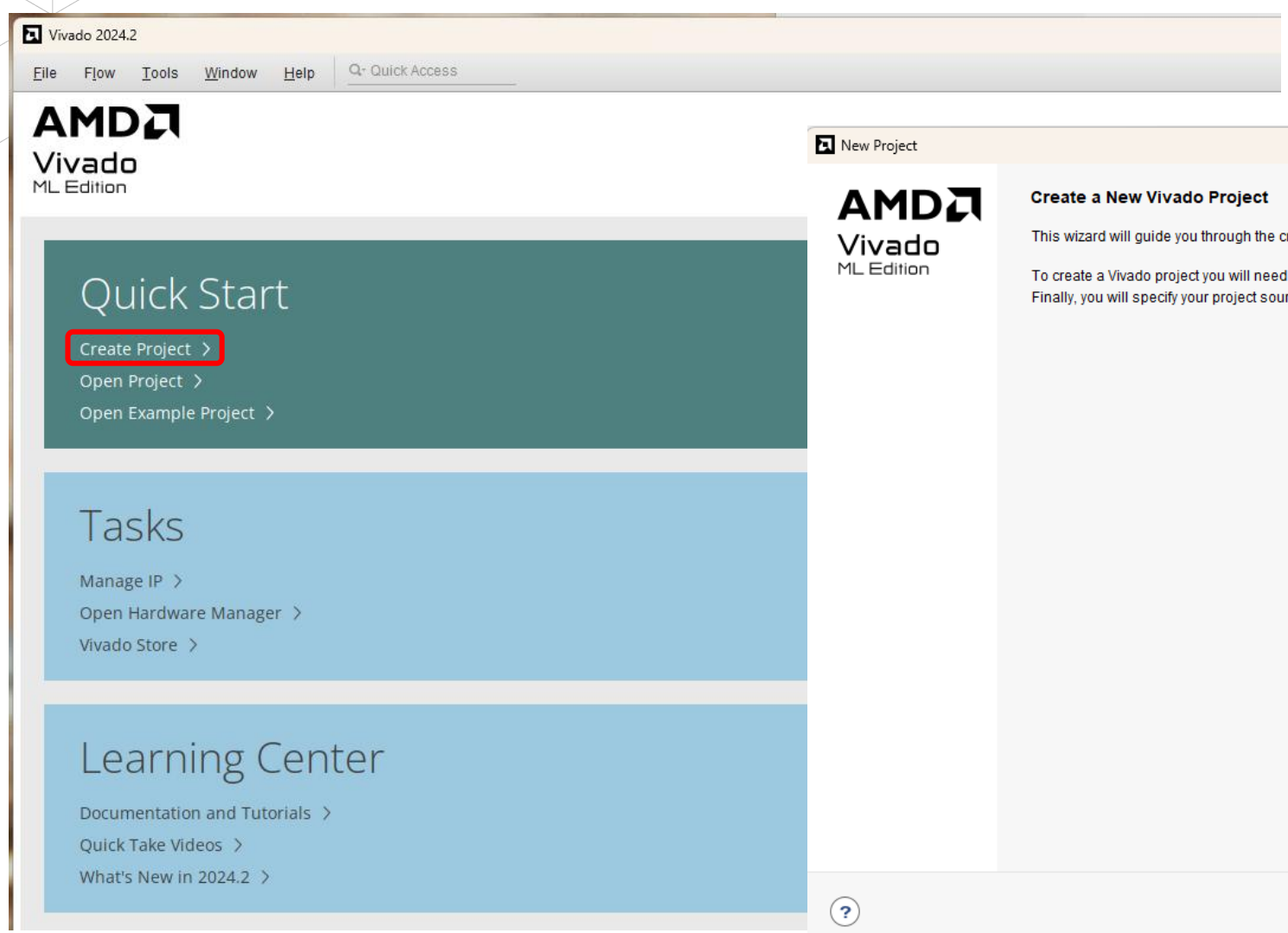
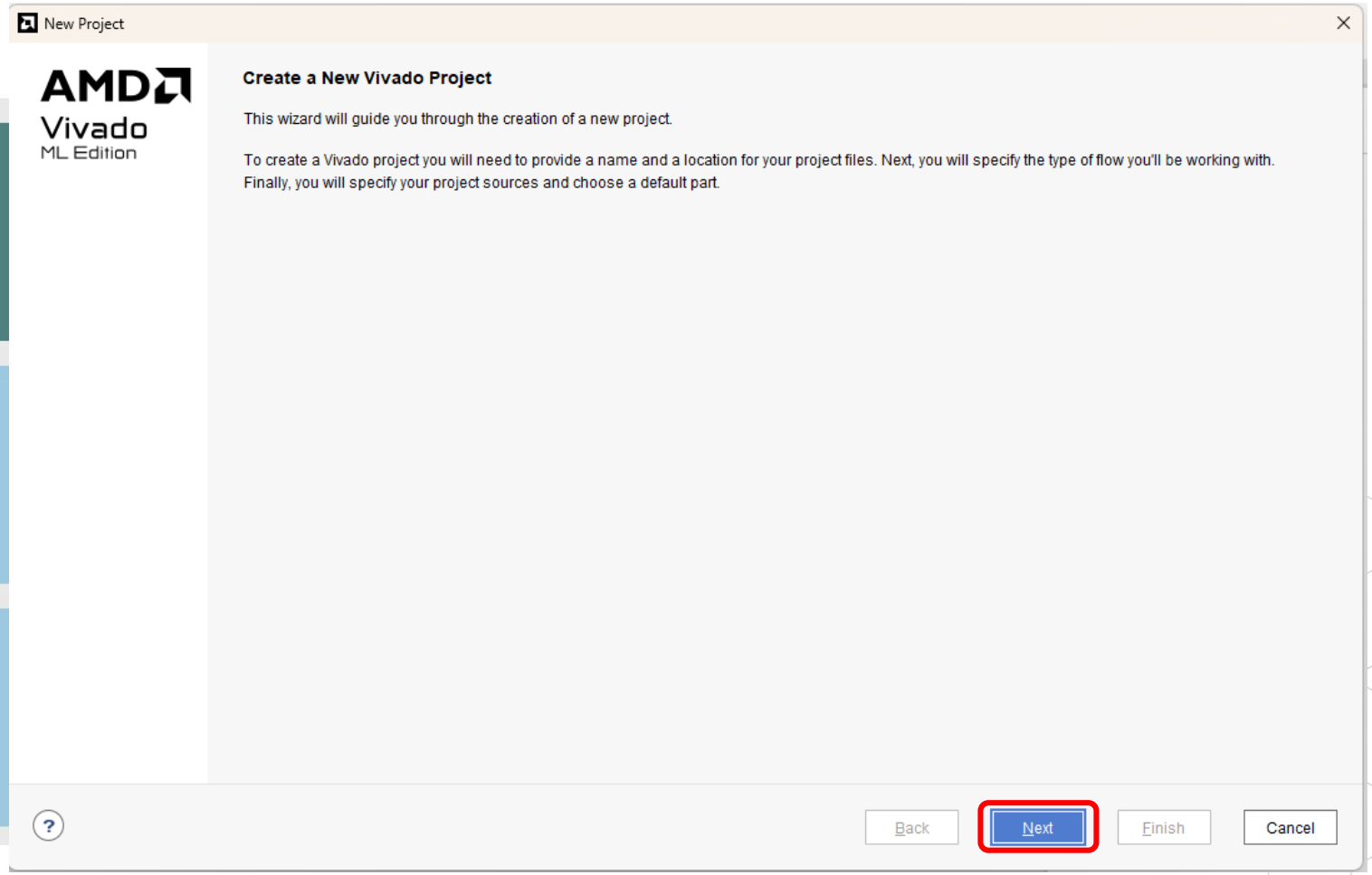
Semiconductor Design Capstone Project | 2026-1

Session3 : Vivado Basic Flow

Vivado Basic Flow





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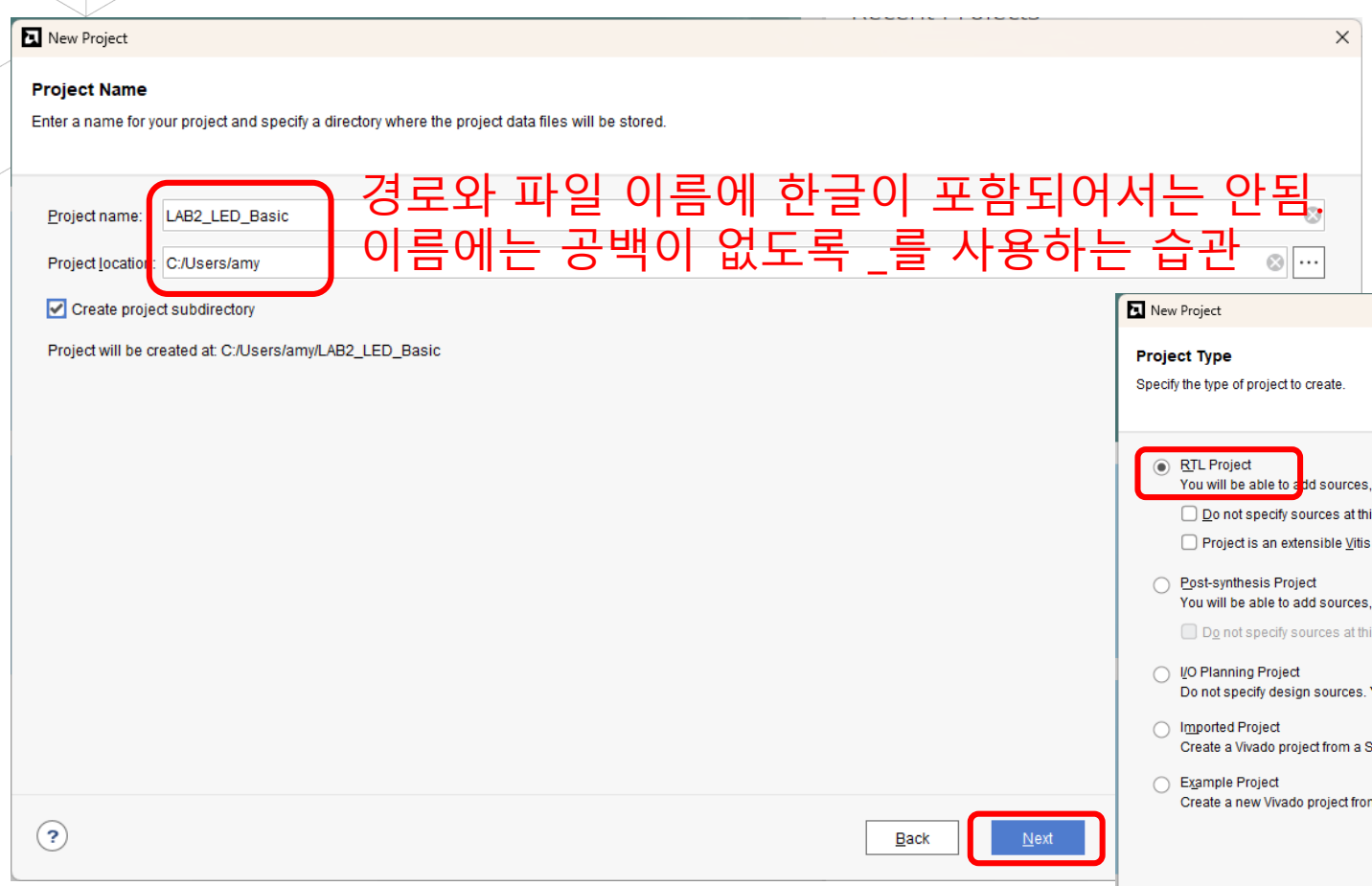
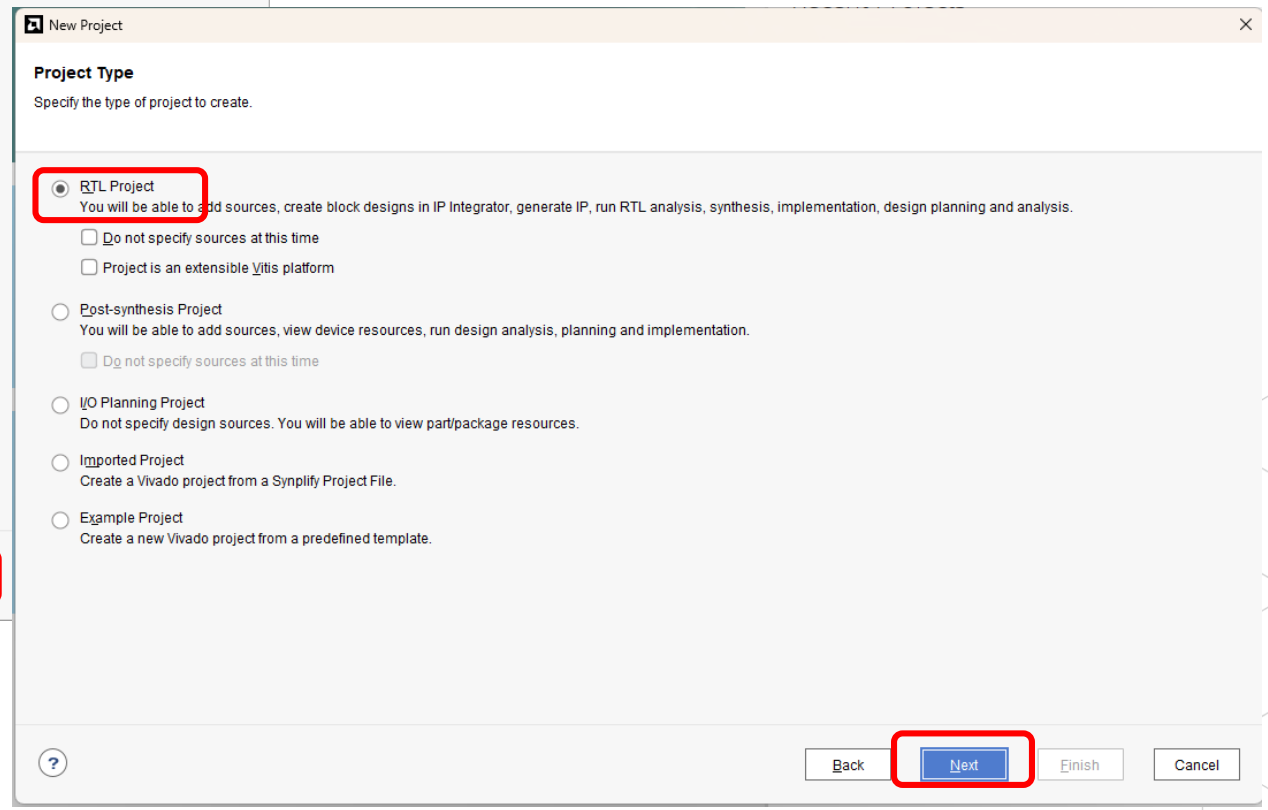
Vivado Basic Flow



경로와 파일 이름에 한글이 포함되어서는 안됨
이름에는 공백이 없도록 _를 사용하는 습관



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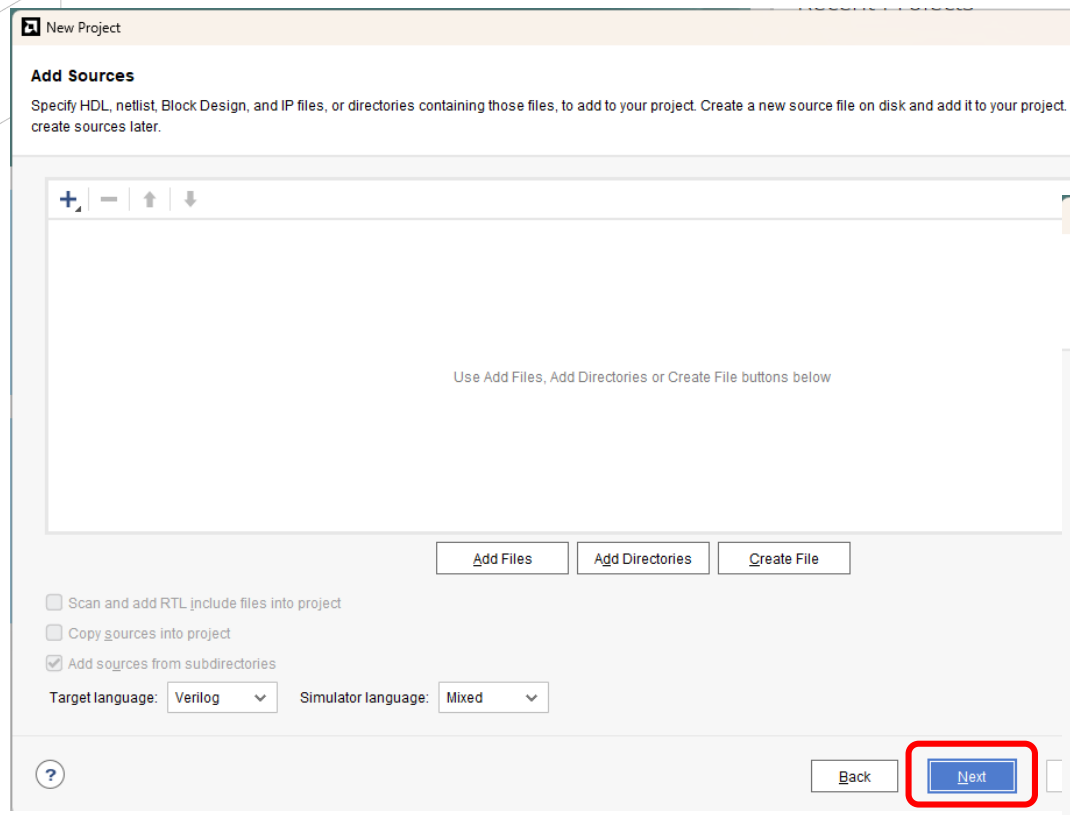
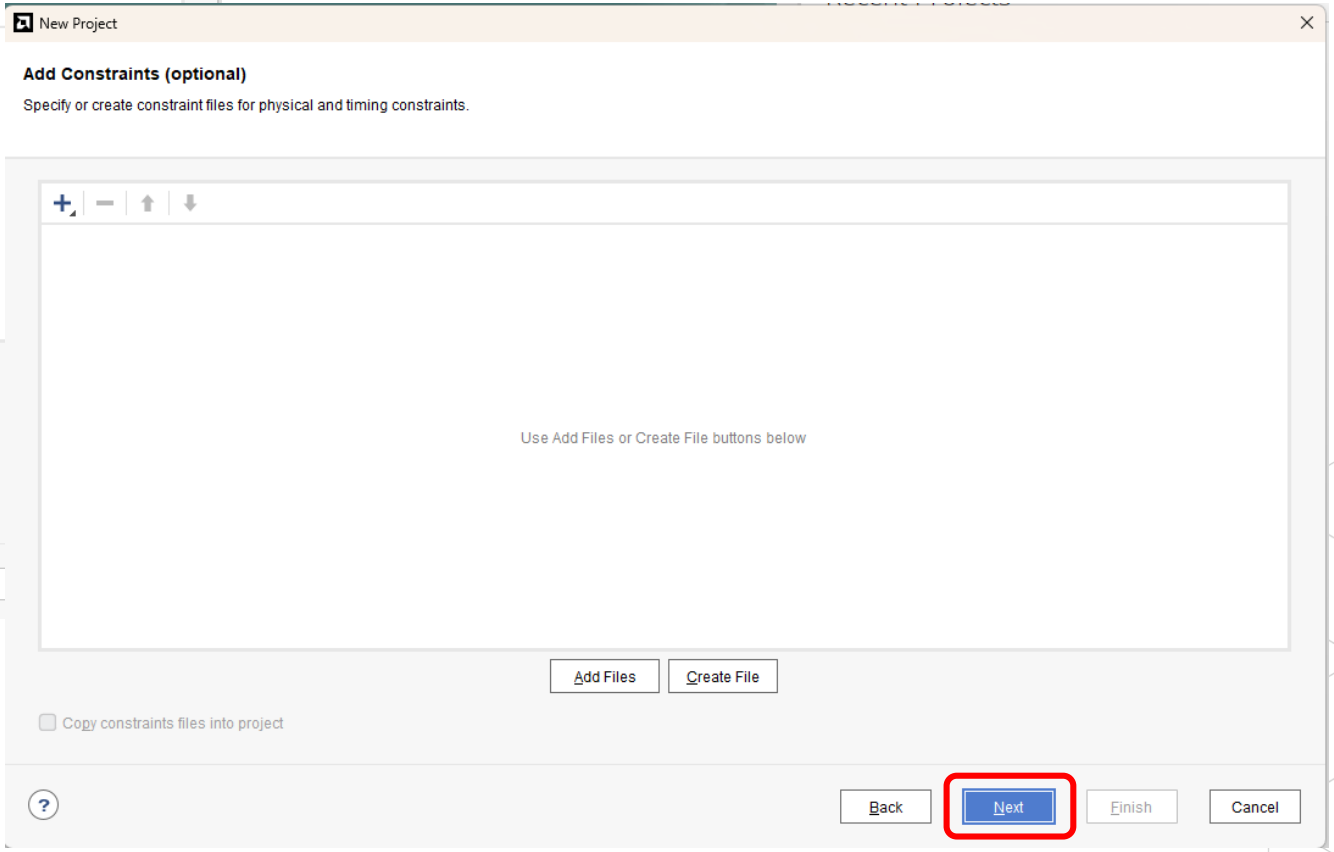
Vivado Basic Flow





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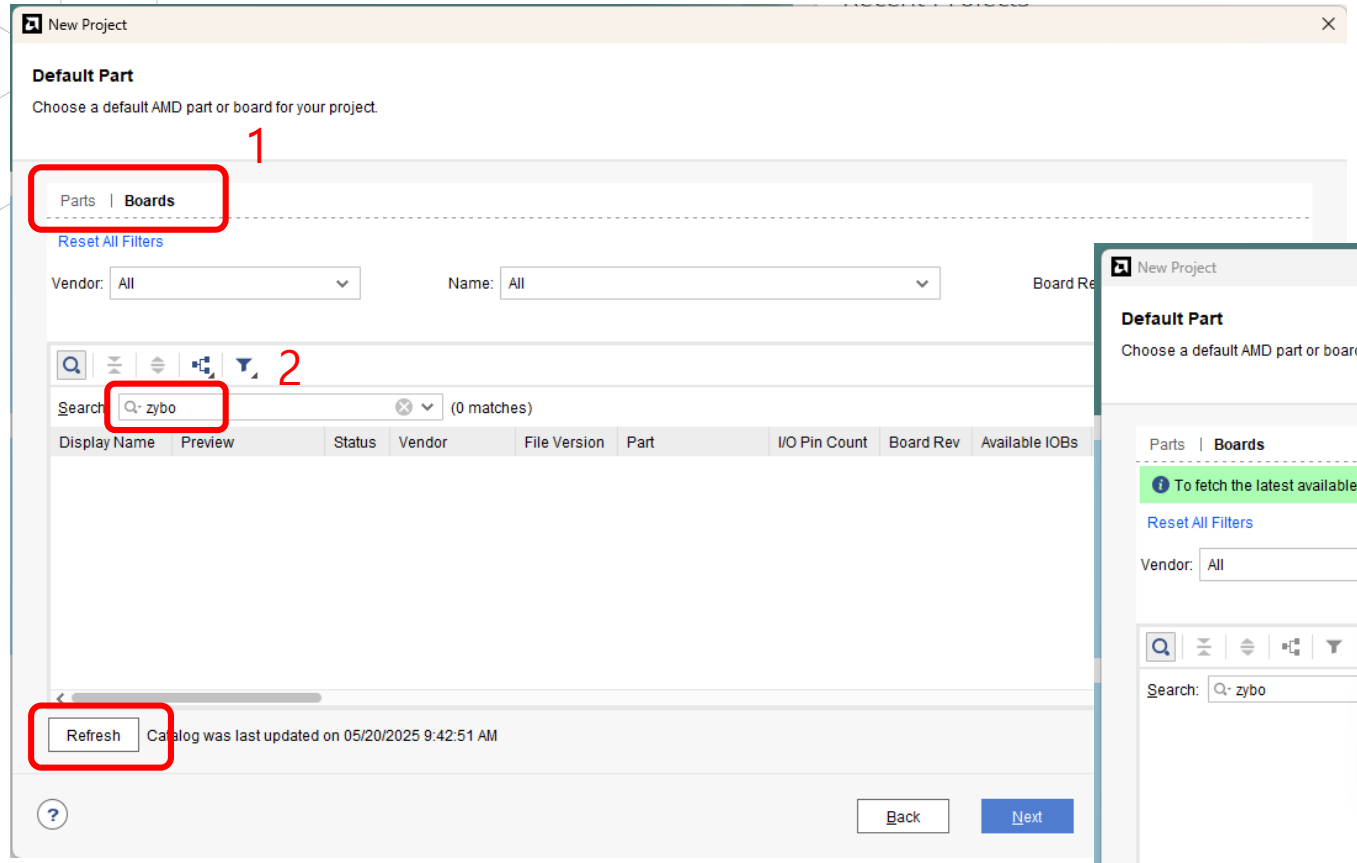
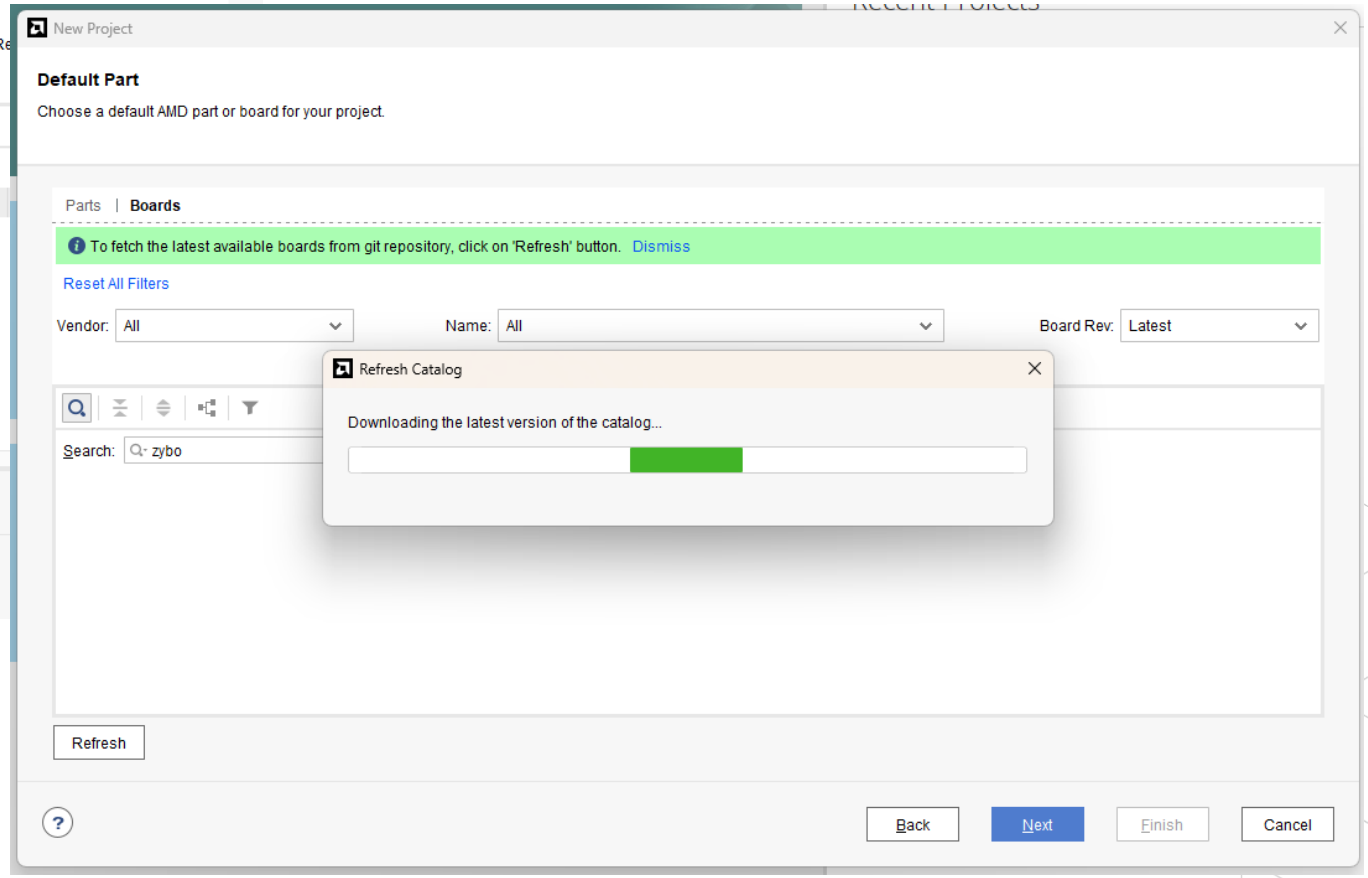
Vivado Basic Flow





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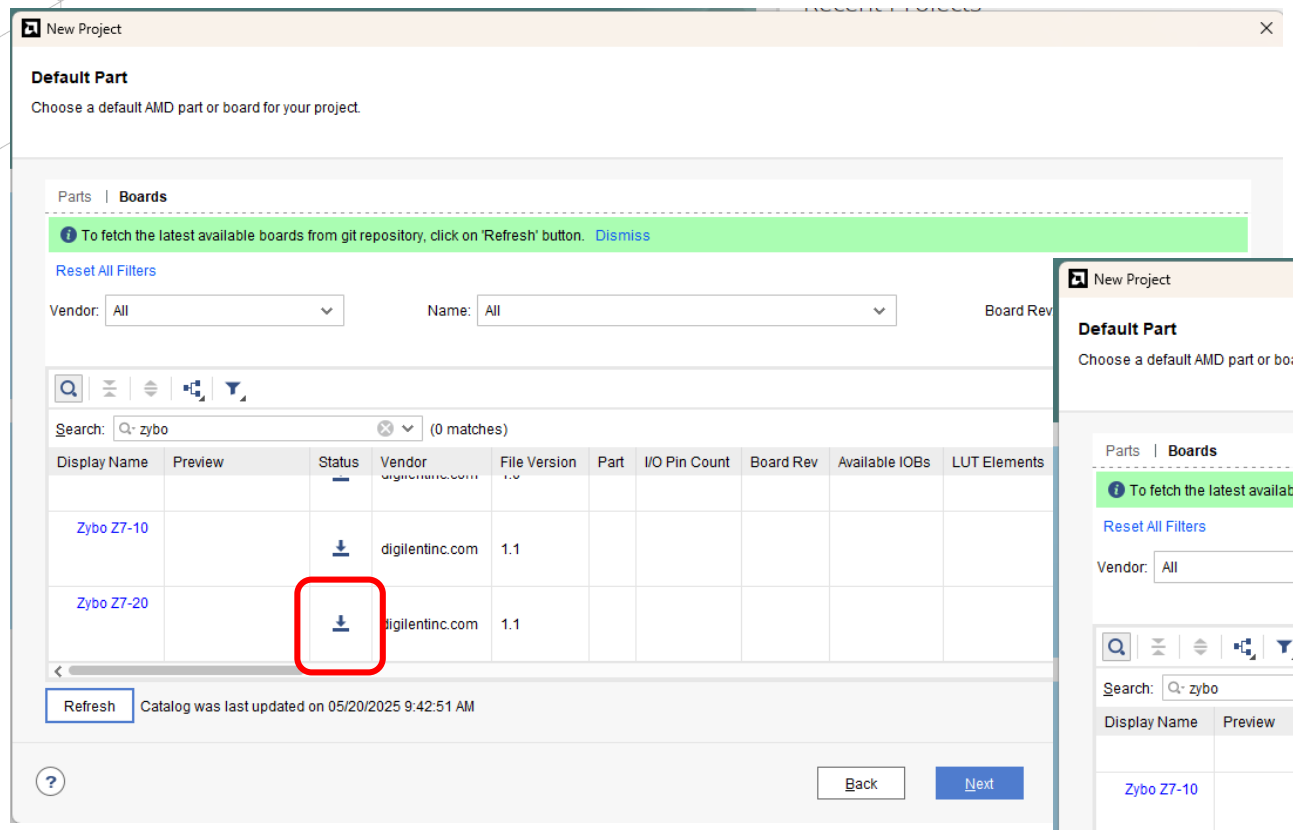
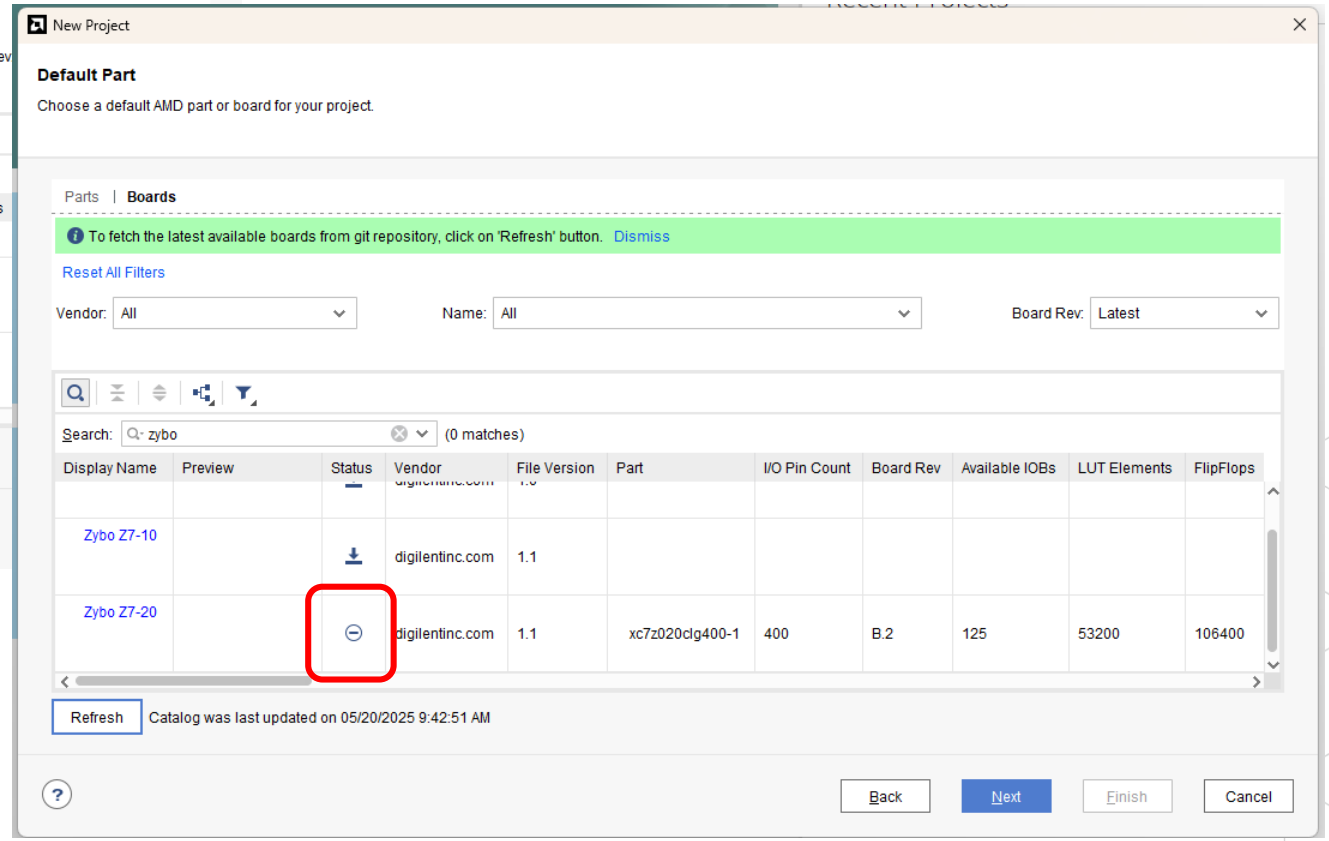
Vivado Basic Flow





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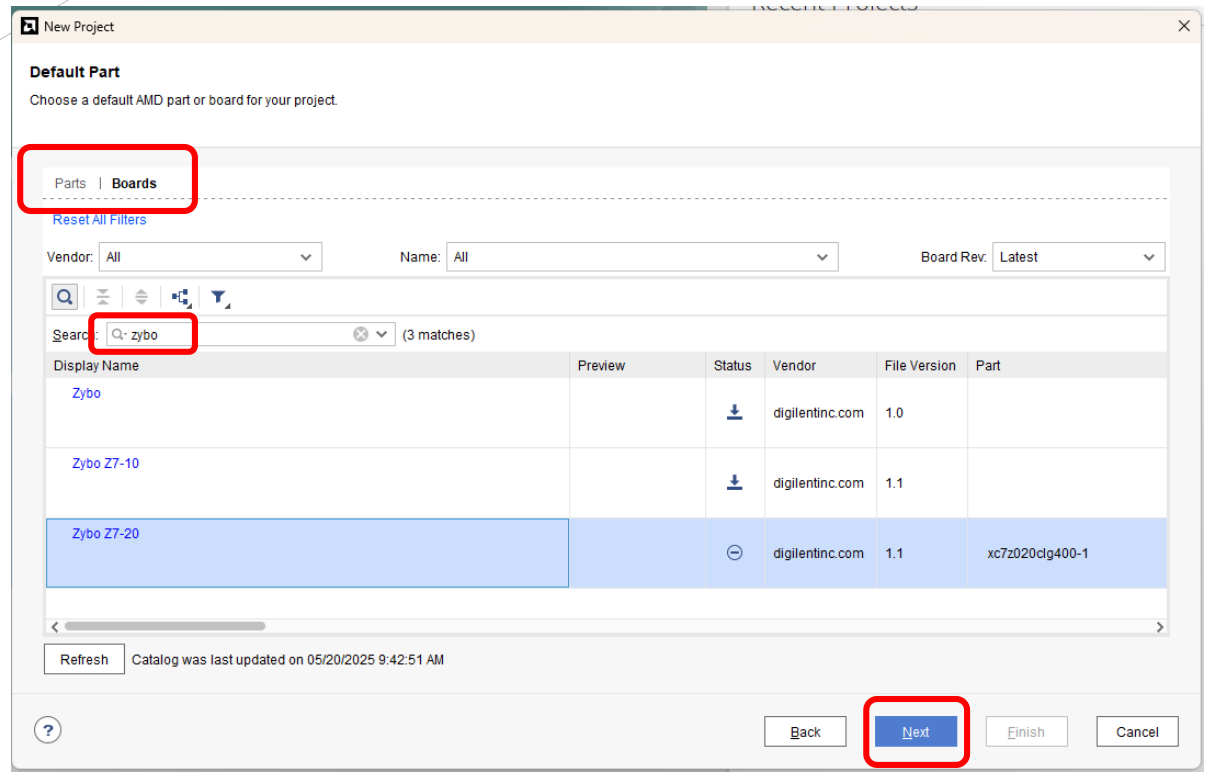
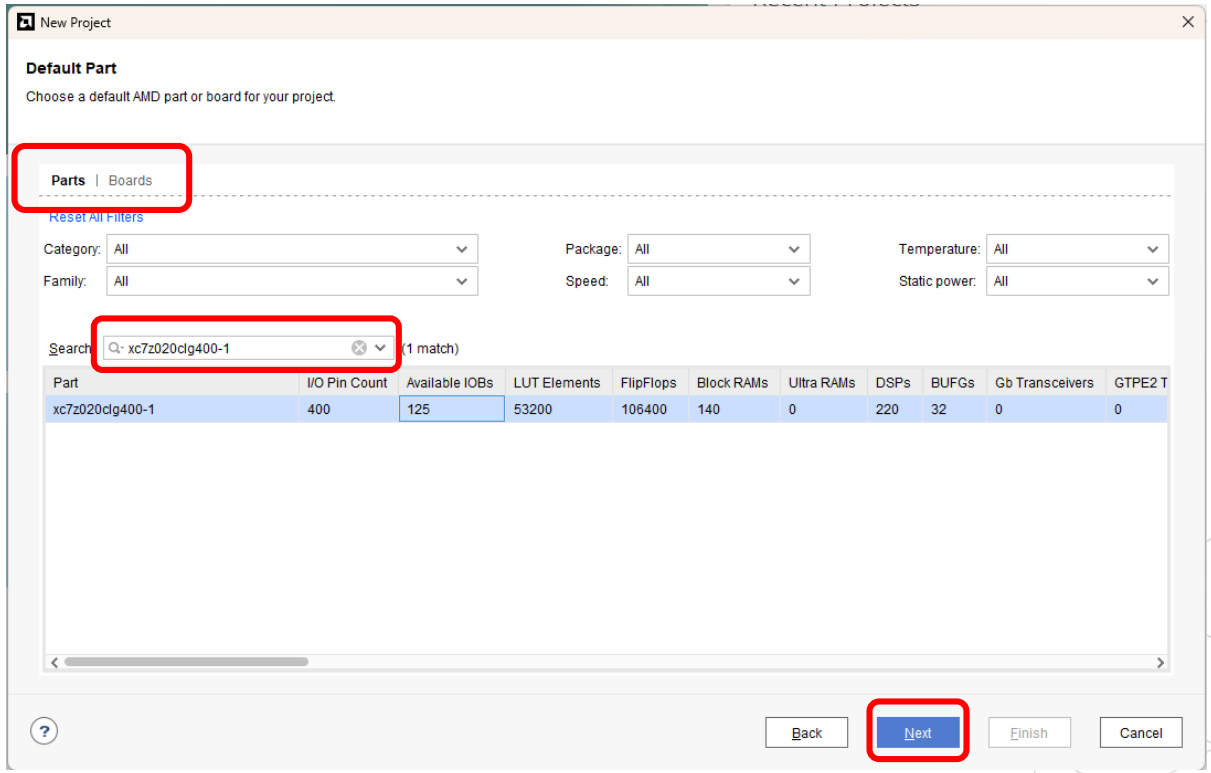
Vivado Basic Flow

둘 중 한가지 방법으로 하면 됩니다.





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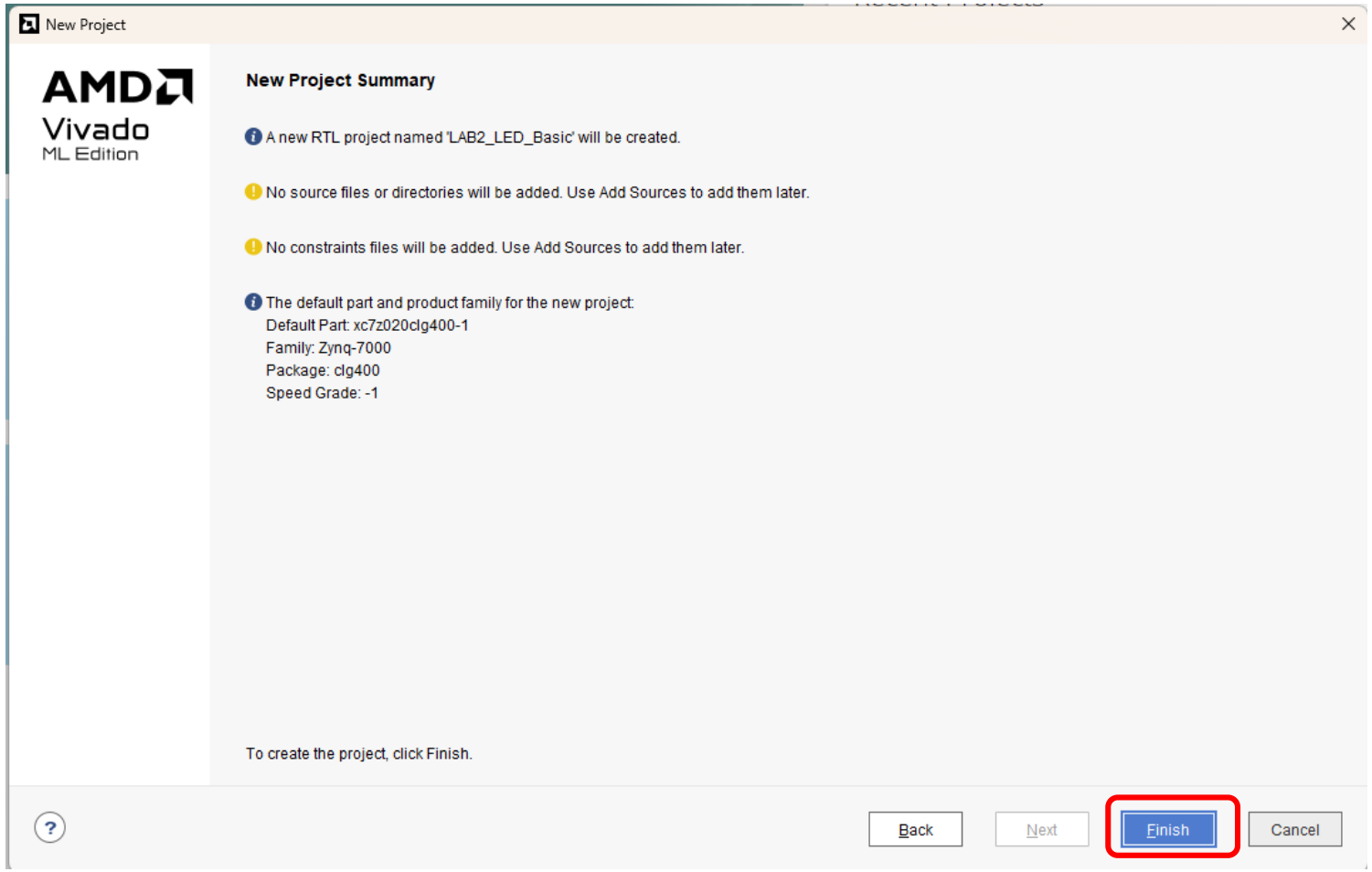
Vivado Basic Flow



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Vivado Basic Flow

LAB2_LED_Basic - [C:/Users/amy/LAB2_LED_Basic/LAB2_LED_Basic.xpr] - Vivado 2024.2

File Edit Flow Tools Reports Window Layout View Help Quick Access

Flow Navigator

PROJECT MANAGER
- Settings
- Add Sources
- Language Templates
- IP Catalog

IP INTEGRATOR
- Create Block Design
- Open Block Design
- Generate Block Design

SIMULATION
- Run Simulation

RTL ANALYSIS
- Run Linter
- Open Elaborated Design

SYNTHESIS
- Run Synthesis
- Open Synthesized Design

IMPLEMENTATION
- Run Implementation
- Open Implemented Design

PROJECT MANAGER - LAB2_LED_Basic

Sources

Design S
Constrair
Simulatio
sim_
Utility Sou

Properties... Ctrl+E
Hierarchy Update
Refresh Hierarchy
IP Hierarchy
Copy Constraints Set...
Edit Constraints Sets...
Edit Simulation Sets...
Add Sources... Alt+A

Hierarchy Libraries Compile Order

Properties

Select an object to see properties

Add Sources

AMD Vivado ML Edition

Add Sources

This guides you through the process of adding and creating sources for your project

- Add or create constraints
- Add or create design sources
- Add or create simulation sources

Back Next Finish Cancel

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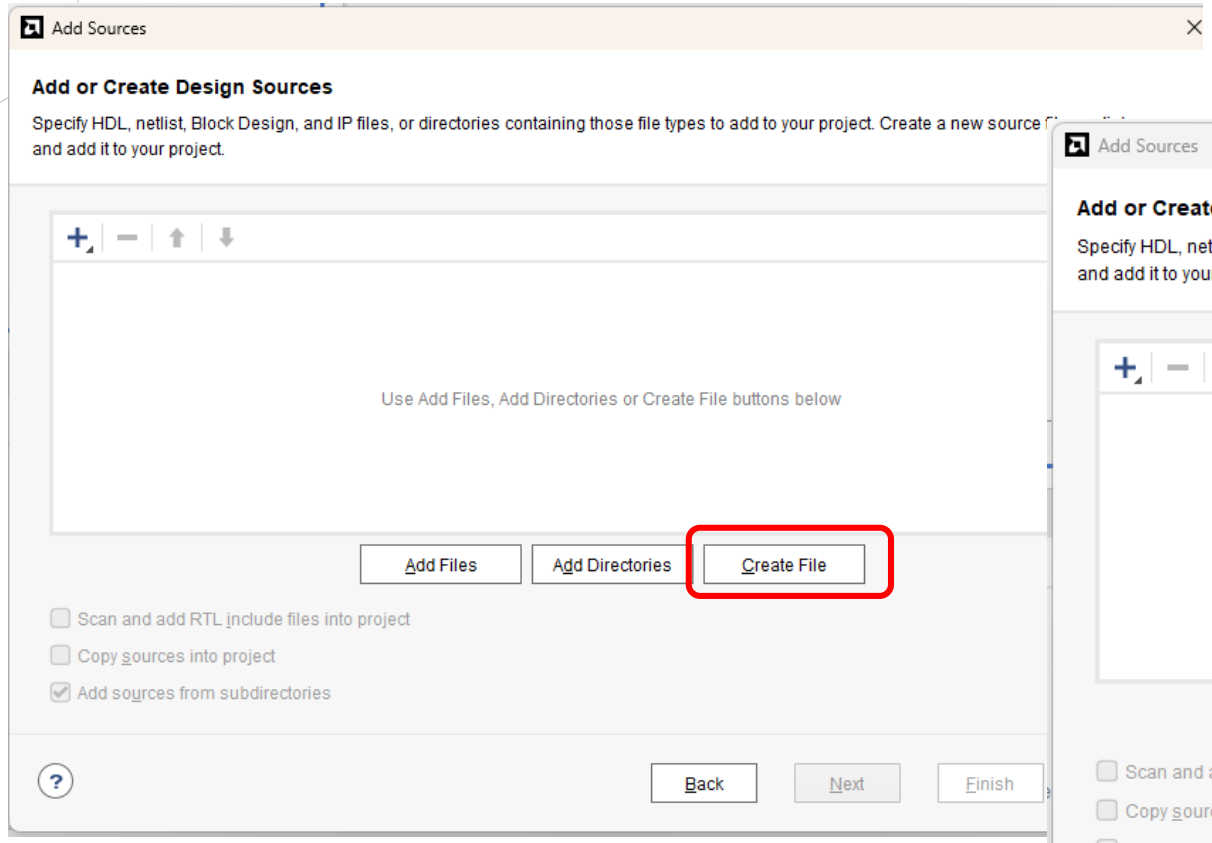
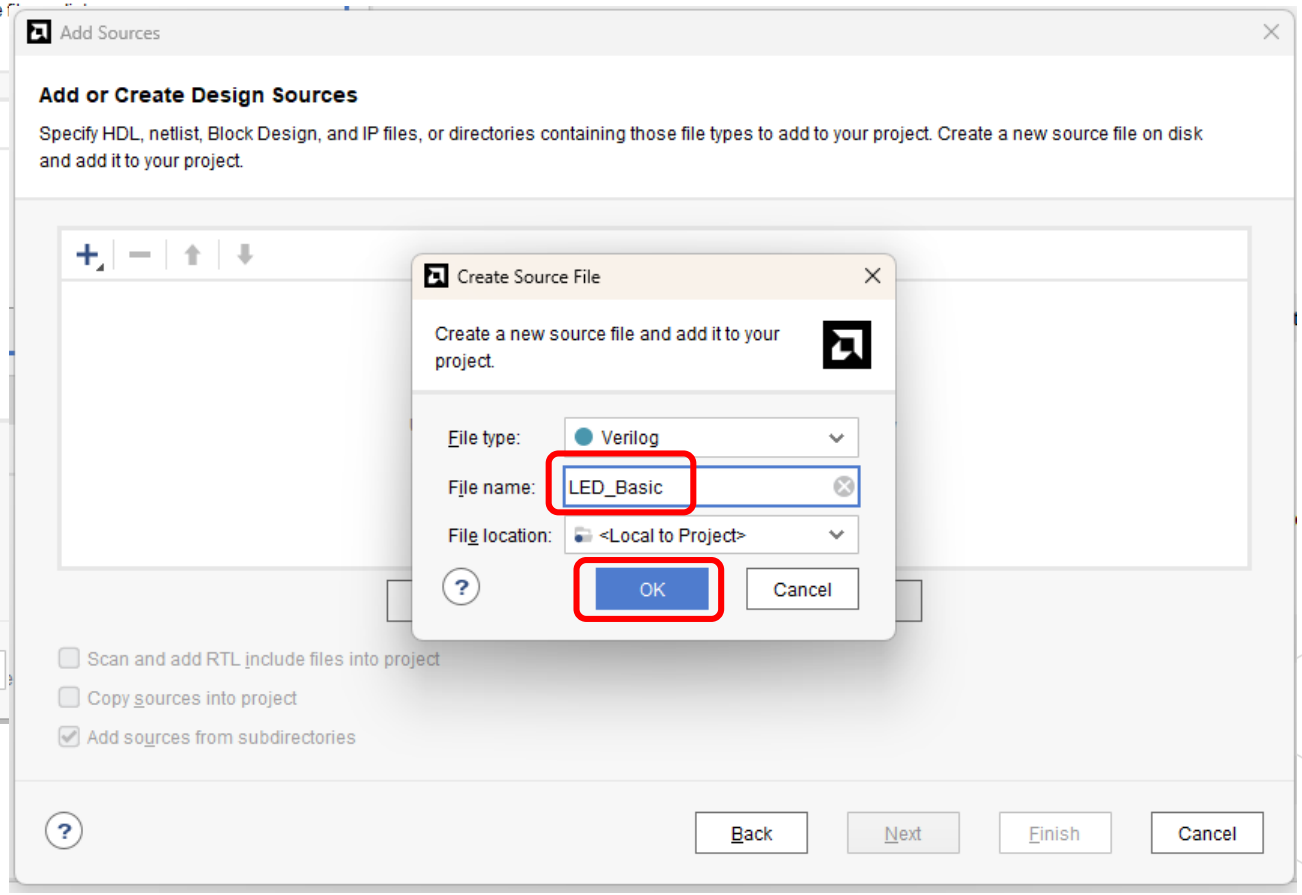
Vivado Basic Flow





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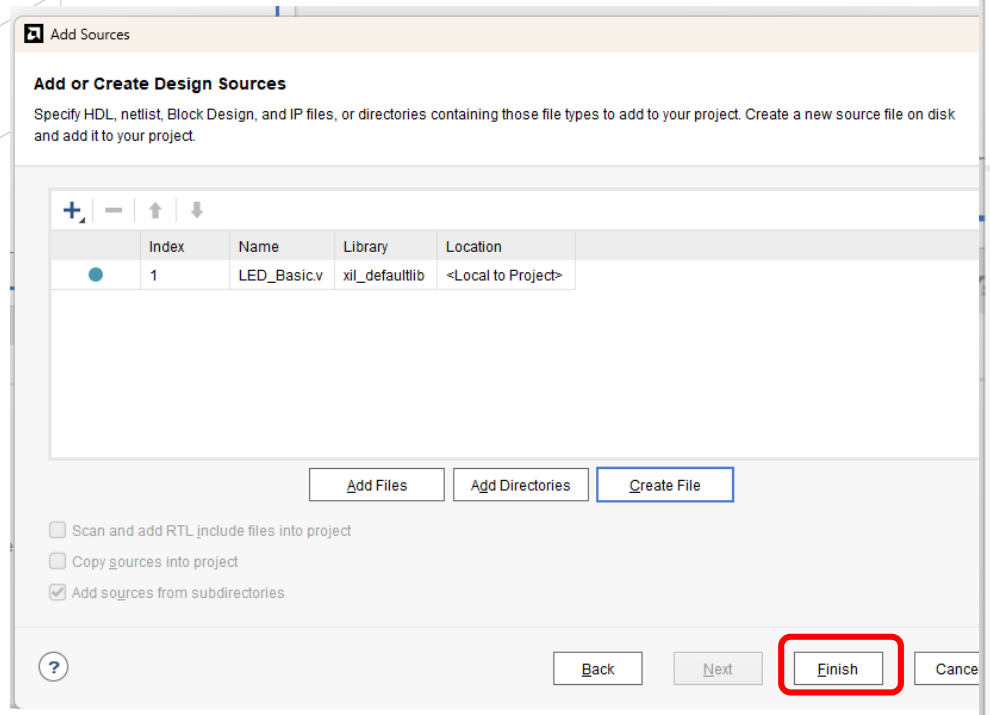

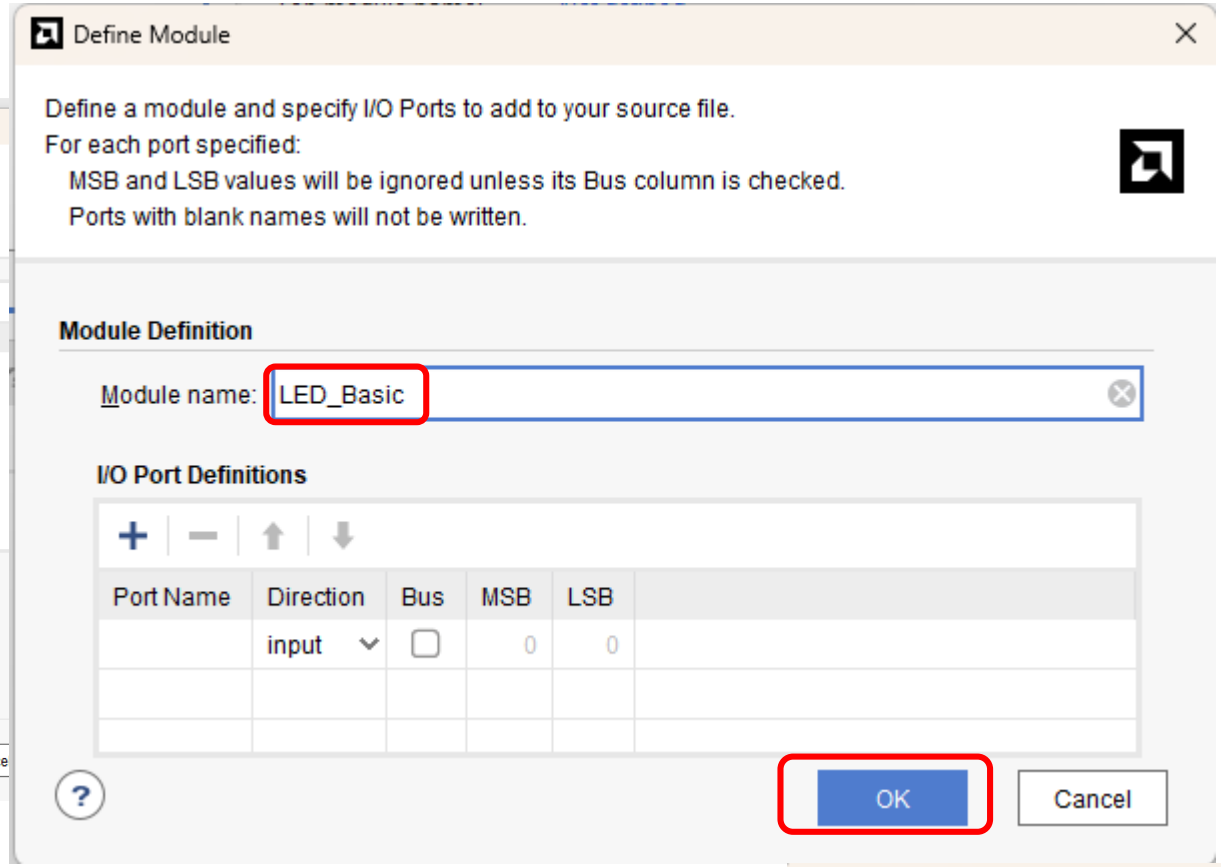

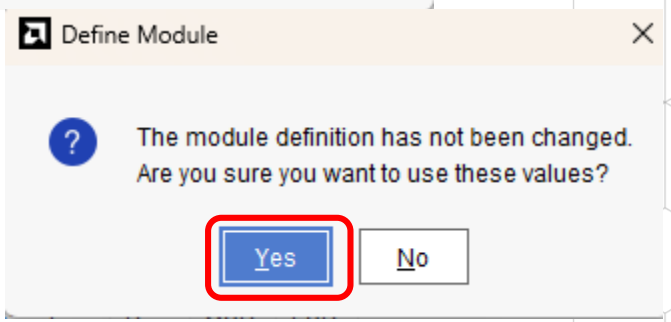

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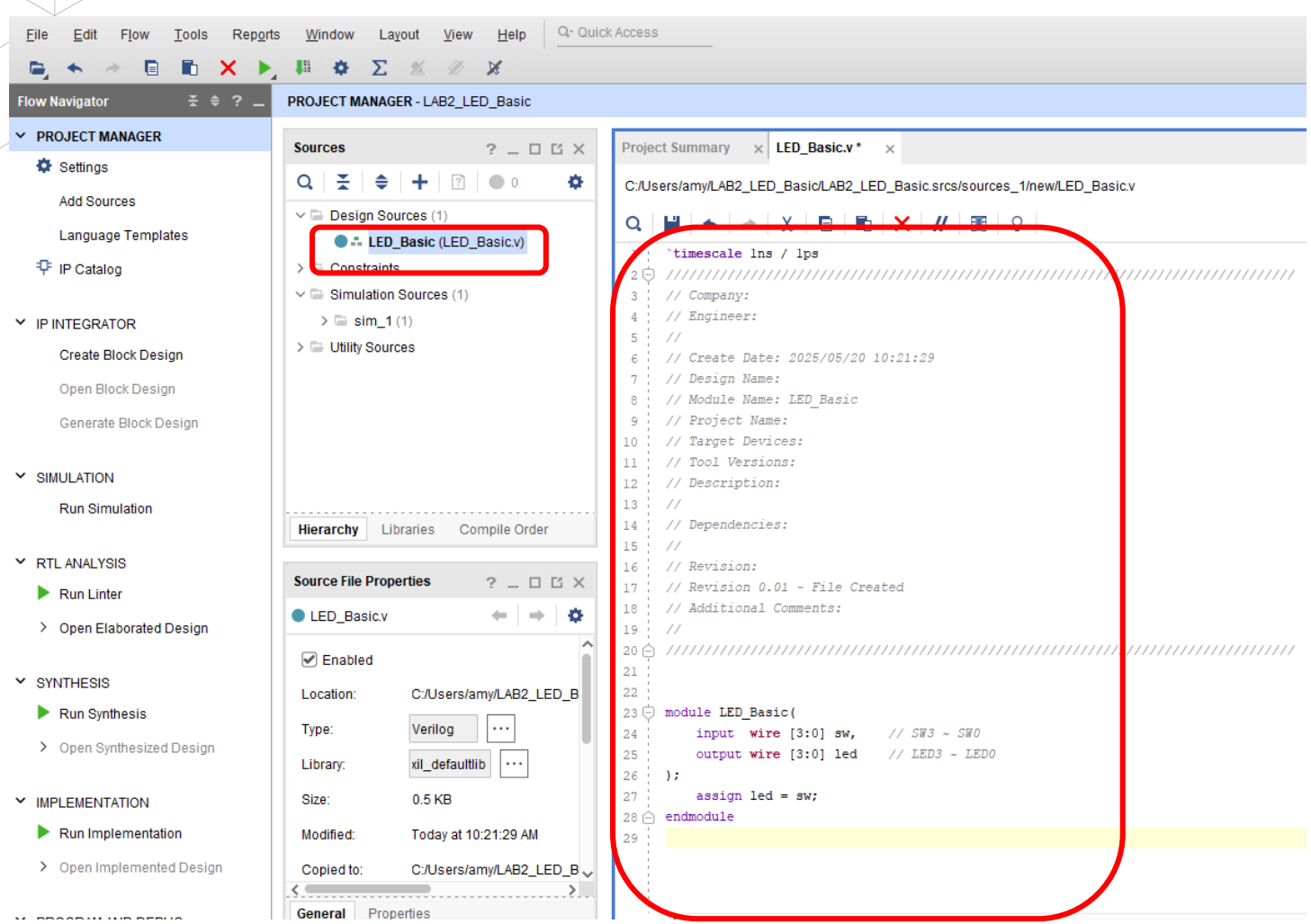
Vivado Basic Flow



복사용 Code

```
module LED_Basic(
    input  wire [3:0] sw,    // SW3 ~ SW0
    output wire [3:0] led    // LED3 ~ LED0
);
    assign led = sw;
endmodule
```

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Vivado Basic Flow

File Edit Flow Tools Reports Window Layout View Help Quick Access Running synth_design Cancel

Default Layout

Flow Navigator

- PROJECT MANAGER
  - Settings
  - Add Sources
  - Language Templates
  - IP Catalog
- IP INTEGRATOR
  - Create Block Design
  - Open Block Design
  - Generate Block Design
- SIMULATION
  - Run Simulation
- RTL ANALYSIS
  - Run Linter
  - Open Elaborated Design
- SYNTHESIS
  - Run Synthesis
  - Open Synthesized Design

PROJECT MANAGER - LAB2_LED_Basic

Sources

- Design Sources (1)
  - LED_Basic (LED_Basic.v)
- Constraints
- Simulation Sources (1)
- Utility Sources

Hierarchy | Libraries | Compile Order

Source File Properties

LED_Basic.v

Enabled

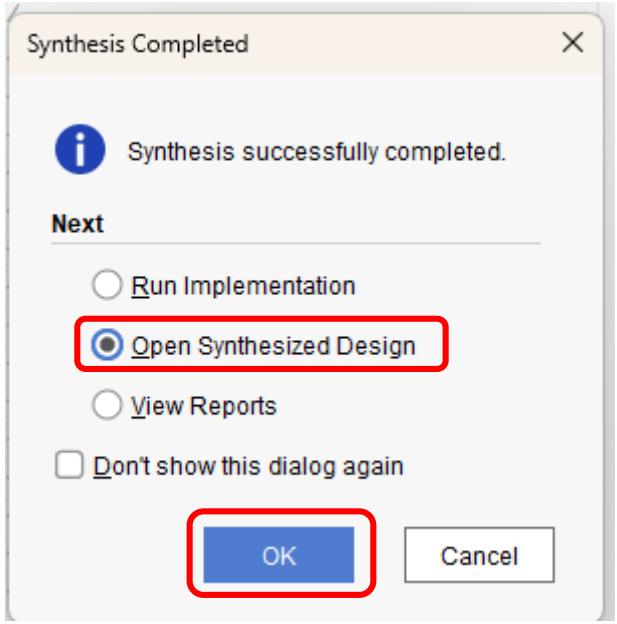
Location: C:/Users/amy/LAB2_LED_B

Type: Verilog

Project Summary | LED_Basic.v

C:/Users/amy/LAB2_LED_Basic/LAB2_LED_Basic.srcs/sour

```
1  `timescale 1ns / 1ps
2  //////////////////////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 2025/05/20 10:21:29
7  // Design Name:
8  // Module Name: LED_Basic
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////////////////////////////////////////////////
21
22
23 module LED_Basic(
24     input  wire [3:0] sw,     // SW3 ~ SW0
```



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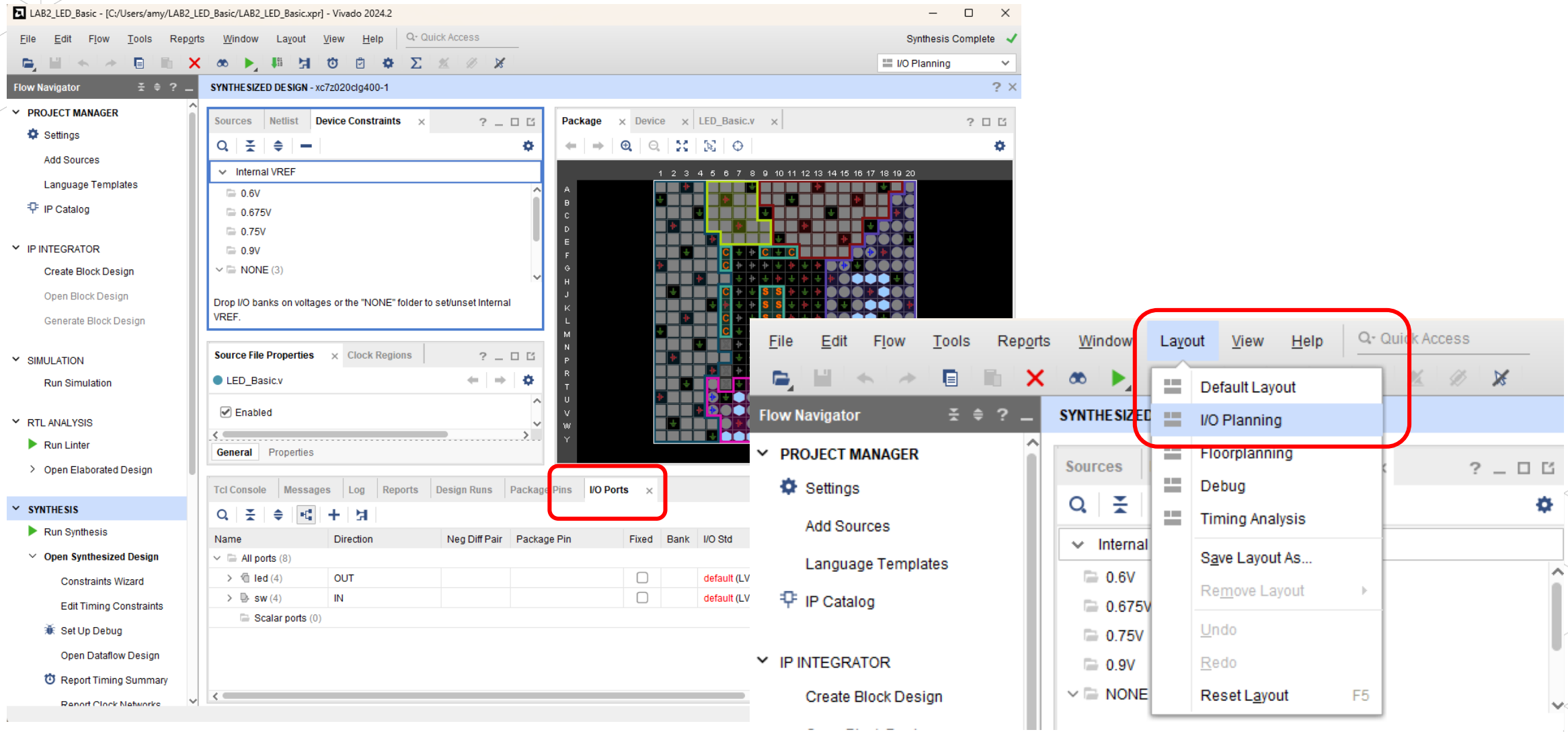
Vivado Basic Flow



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Vivado Basic Flow

Control + S

Tcl Console | Messages | Log | Reports | Design Runs | Package Pins | I/O Ports

Name	Direction	Neg Diff Pair	Package Pin	Fixed	Bank	I/O Std	Vcco	Vref
All ports (8)								
led (4)	OUT			☑	35	LVCMOS33*	3.300	
led[3]	OUT		D18	☑	35	LVCMOS33*	3.300	
led[2]	OUT		G14	☑	35	LVCMOS33*	3.300	
led[1]	OUT		M15	☑	35	LVCMOS33*	3.300	12
led[0]	OUT		M14	☑	35	LVCMOS33*	3.300	12
sw (4)	IN			☑	(Multi)	LVCMOS33*	3.300	
sw[3]	IN		T16	☑	34	LVCMOS33*	3.300	
sw[2]	IN		W13	☑	34	LVCMOS33*	3.300	
sw[1]	IN		P15	☑	34	LVCMOS33*	3.300	
sw[0]	IN		G15	☑	35	LVCMOS33*	3.300	
Scalar ports (0)								

I/O Bank 502

Out of Date Design

Saving the current constraints to the target project constraints file may cause your synthesis to go out-of-date. To avoid re-running synthesis, you can force the design up-to-date by selecting the run in the Design Runs tab, right clicking, and selecting 'Force Up-to-Date'.

☐ Don't show this dialog again

OK

Save Constraints

Select a target file to write new unsaved constraints to. Choosing an existing file will update that file with the new constraints.

◉ Create a new file

File type: XDC

File name: constraint_LED_Basic

File location: <Local to Project>

○ Select an existing file

<select a target file>

OK | Cancel

https://github.com/Digilent/Zybo-Z7-20-XADC/blob/master/src/constraints/Zybo-Z7.xdc

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Vivado Basic Flow

File Edit Flow Tools Reports Window Layout View Help

Flow Navigator

SIMULATION
- Run Simulation

RTL ANALYSIS
- Run Linter
- Open Elaborated Design

SYNTHESIS
- Run Synthesis
- Open Synthesized Design
  - Constraints Wizard
  - Edit Timing Constraints
  - Set Up Debug
  - Open Dataflow Design
  - Report Timing Summary
  - Report Clock Networks
  - Report Clock Interaction
  - Report Methodology
  - Report DRC
  - Report Noise
  - Report Utilization
  - Report Power
  - Schematic

IMPLEMENTATION
- Run Implementation
- Open Implemented Design

PROGRAM AND DEBUG
- Generate Bitstream
- Open Hardware Manager

SYNTHESIZED DESIGN - xc7z020clg400-1

Sources | Netlist | Device Constraints

Design Sources (1)
- LED_Basic (LED_Basic.v)

Constraints (1)

Hierarchy | Libraries | Compile Order

I/O Port Bus Properties | Clock Regions

sw

General | I/O Ports | Magnify | Power

Package | Device | LED_Basic.v

C:/Users/amy/LAB2_LED_Basic/LAB2_LED_Basic.srcs/sources_1/new/LED_Basic.v

```
19  //
20  ////////////////////////////////////////////////////////////////////////////////
21
22
23  module LED_Basic(
24      input  wire [3:0] sw,     // SW3 ~ SW0
25      output wire [3:0] led     // LED3 ~ LED0
26  );
27      assign led = sw;
28  endmodule
29
```

Tcl Console | Messages | Log | Reports | Design Runs | Package Pins | I/O Ports

Name	Direction	Neg Diff Pair	Package Pin	Fixed	Bank	I/O Std	Vcco	Vref	Drive Strength	Slew Type
All ports (8)										
led (4)	OUT			☑	35	LVCMOS33*	3.300		12	
led[3]	OUT		D18	☑	35	LVCMOS33*	3.300		12	
led[2]	OUT		G14	☑	35	LVCMOS33*	3.300		12	
led[1]	OUT		M15	☑	35	LVCMOS33*	3.300		12	
led[0]	OUT		M14	☑	35	LVCMOS33*	3.300		12	
sw (4)	IN			☑	(Multipl	LVCMOS33*	3.300			
sw[3]	IN		T16	☑	34	LVCMOS33*	3.300			
sw[2]	IN		W13	☑	34	LVCMOS33*	3.300			
sw[1]	IN		P15	☑	34	LVCMOS33*	3.300			
sw[0]	IN		G15	☑	35	LVCMOS33*	3.300			
Scalar ports (0)										

I/O Bank 502

No Implementation Results Available

There are no implementation results available. OK to launch synthesis and implementation?
'Generate Bitstream' will automatically start when synthesis and implementation completes.

☐ Don't show this dialog again

Yes | No

Launch Runs

Launch the selected synthesis or implementation runs and generate bitstream.

Launch directory: <Default Launch Directory>

Options

◉ Launch runs on local host: Number of jobs: 6

○ Generate scripts only

☐ Don't show this dialog again

OK | Cancel

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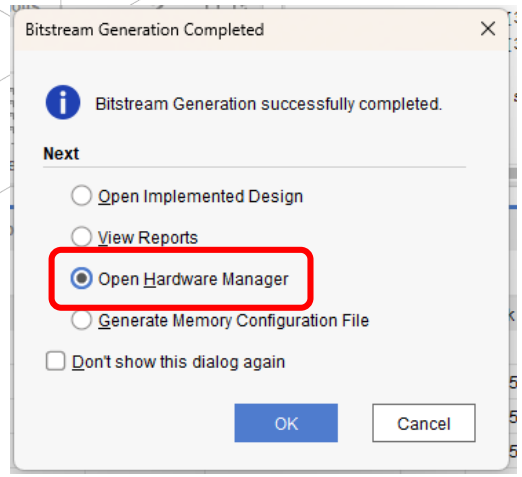
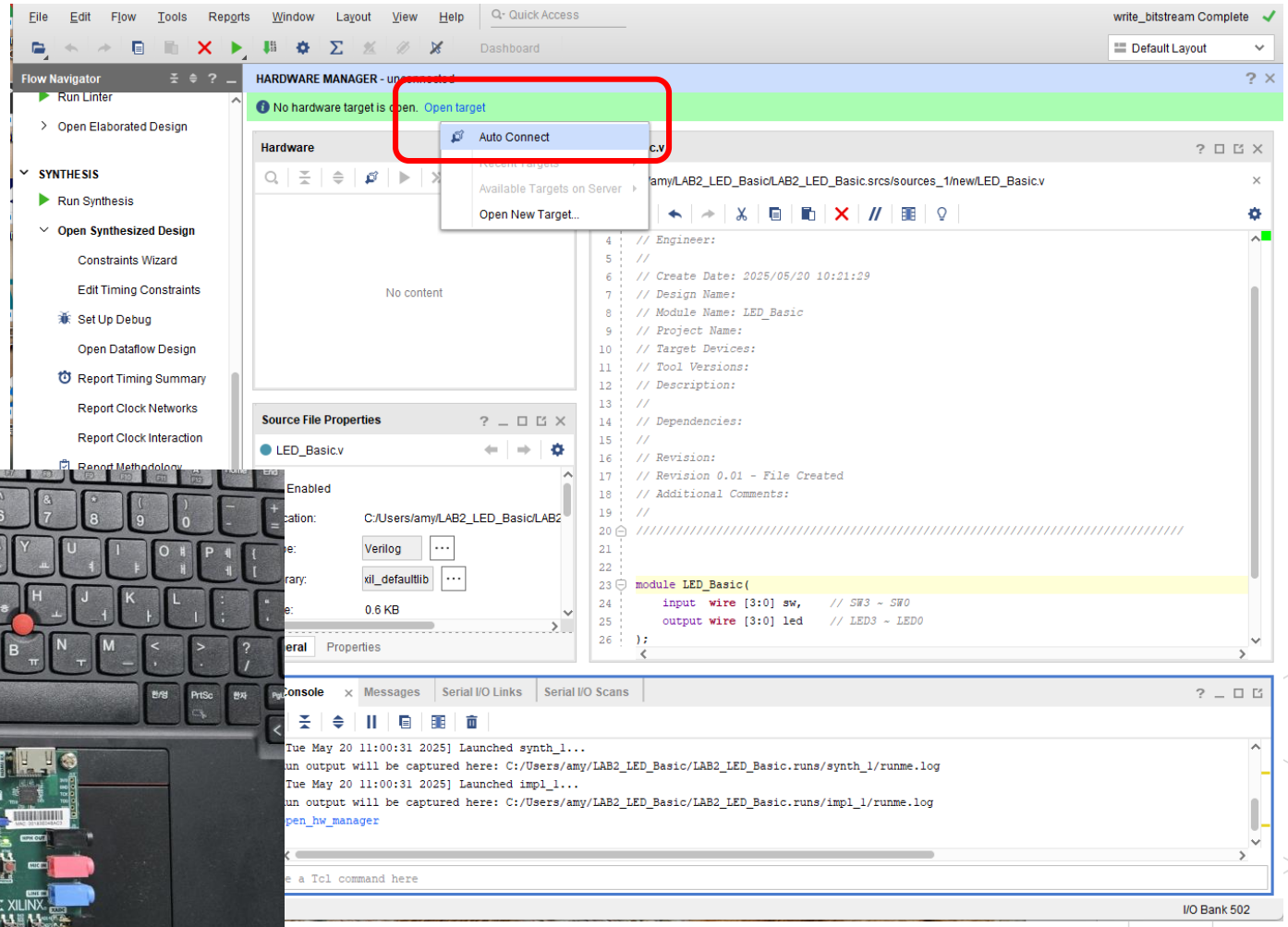
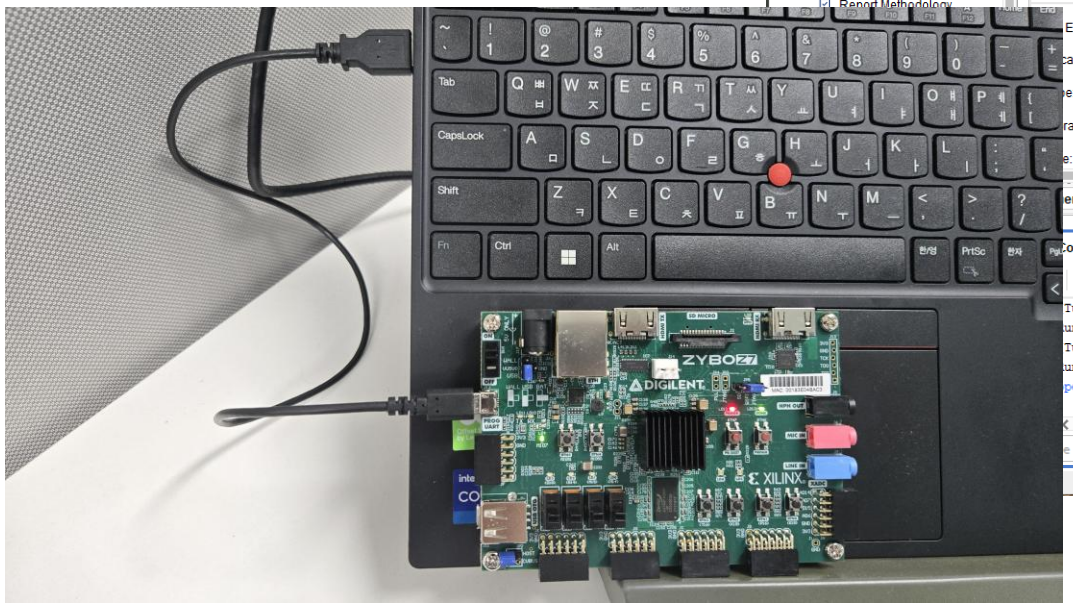
Vivado Basic Flow







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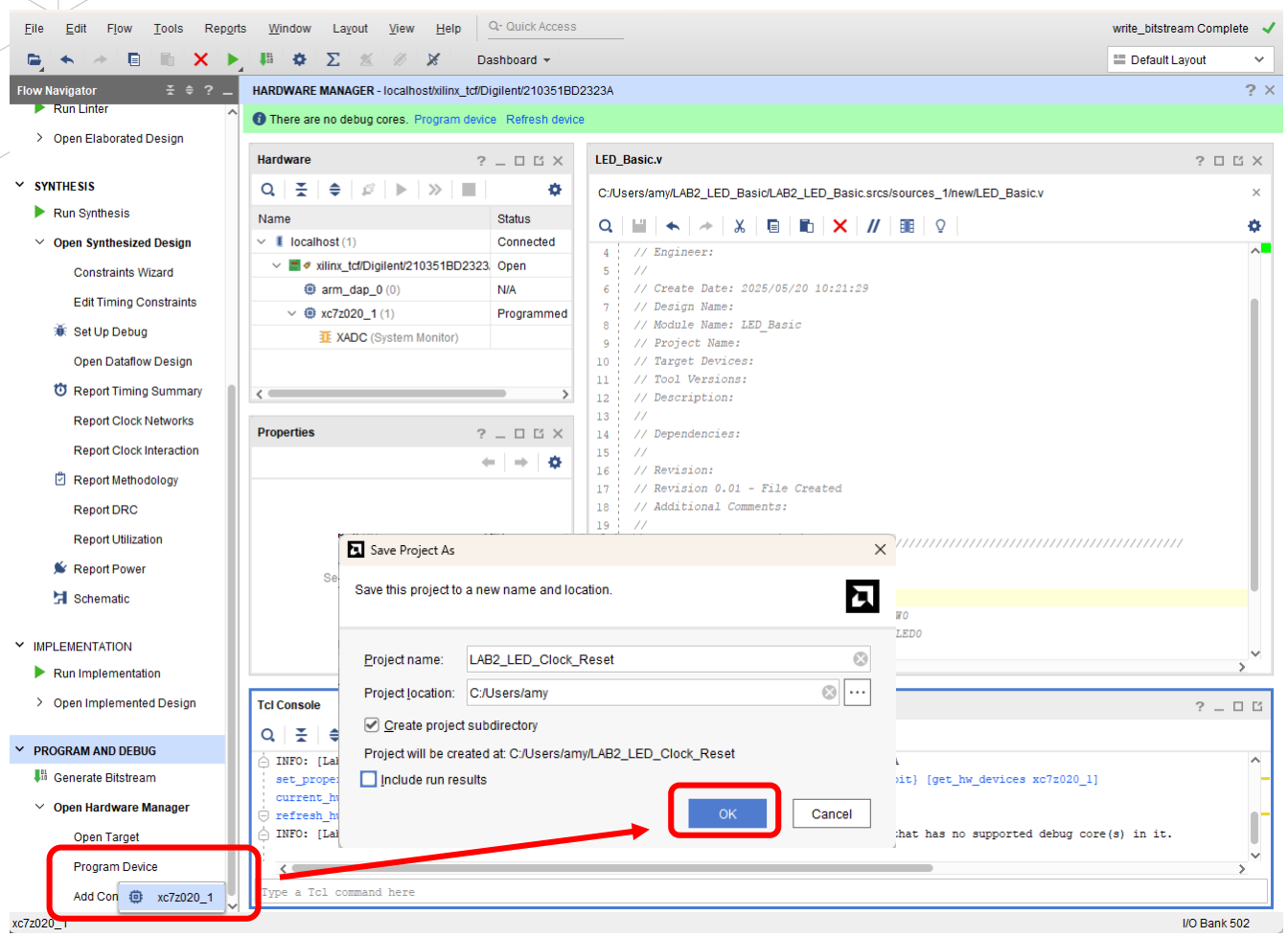
Vivado Basic Flow



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